# Individual Printer Performance and Anomaly Report

## EPR-001

Offset Press A · HeidelPress XL-106 · Plant 1 - Hall A

## PRINTER INFORMATION

| | |
|---|---|
| **Printer ID** | EPR-001 |
| **Printer name** | Offset Press A |
| **Model** | HeidelPress XL-106 |
| **Machine type** | Offset |
| **Location** | Plant 1 - Hall A |
| **Installed** | 2019-01-10 |
| **Last recorded status** | IDLE |
| **Status mix over period** | IDLE 415, PRINTING 424, MAINTENANCE 12, ERROR 13 |
| **Total telemetry readings** | 864 |
| **Monitoring period** | 2026-10-15 00:00:00 to 2026-10-17 23:55:00 |
| **Report generated** | 2026-10-26 13:36:33.480 |

## CONFIGURED SENSOR THRESHOLD LIMITS

| SENSOR | LOWER LIMIT | UPPER LIMIT | SEVERITY |
|---|---|---|---|
| Temperature (°C) | — | 75.0 | CRITICAL |
| Humidity (%) | 30.0 | 70.0 | WARNING |
| Ink level (%) | 15.0 | — | WARNING |
| Paper stock (%) | 10.0 | — | INFO |
| Motor RPM (rpm) | — | 13500.0 | WARNING |
| Vibration (mm/s) | — | 5.5 | CRITICAL |
| Power (kW) | — | 51.3 | WARNING |
| Paper jams (count) | — | 2.0 | CRITICAL |

## SENSOR PERFORMANCE SUMMARY

| SENSOR (UNIT) | MIN | AVG | MAX | LAST VALUE | BREACH READINGS |
|---|---|---|---|---|---|
| Temperature (°C) | 38.7 | 45.92 | 53.5 | 39.4 | 0 |
| Humidity (%) | 36.9 | 49.65 | 85.0 | 53.1 | 40 |
| Ink level (%) | 17.8 | 36.99 | 79.6 | 17.8 | 0 |
| Paper stock (%) | 22.1 | 41.00 | 62.9 | 22.1 | 0 |
| Motor RPM (rpm) | 0.0 | 5473.73 | 11439.0 | 195.0 | 0 |
| Vibration (mm/s) | 0.1 | 1.18 | 2.4 | 0.2 | 0 |
| Power (kW) | 3.0 | 21.73 | 66.3 | 4.0 | 14 |
| Throughput (ppm) | 0.0 | 76.81 | 168.0 | 0.0 | 0 |
| Paper jams (count) | 0.0 | 0.05 | 5.0 | 0.0 | 8 |

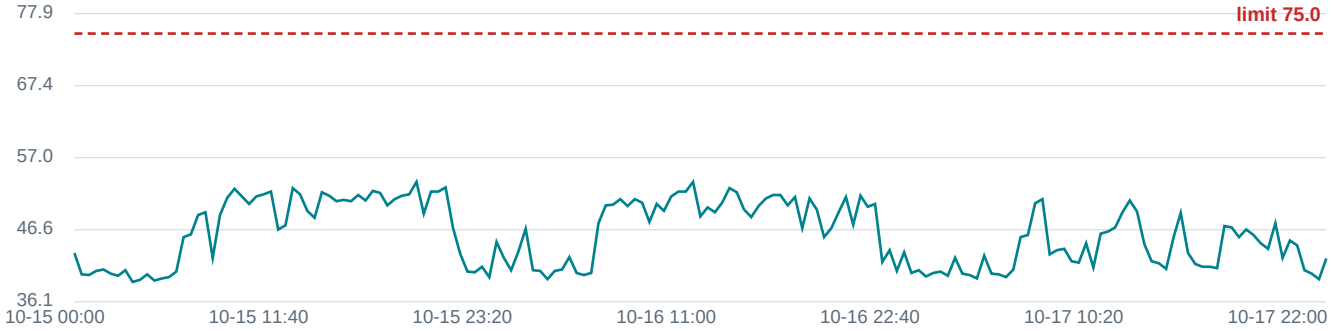
## ANOMALY DETECTION SUMMARY

| | |
|---|---|
| **Labelled anomalous readings** | 75 of 864 readings (8.7%) |
| **Raw threshold-breach readings** | 62 readings with at least one sensor out of bounds |
| **Anomaly types** | INK_LEAK_DROP: 8 · PAPER_JAM_BURST: 13 · POWER_SURGE: 14 · HUMIDITY_SPIKE: 40 |
| **Detected anomaly events** | 12 fired by the live detector (window=3 consecutive readings), last at 2026-10-15 05:55:00 |
| **Alerts in alerts.csv** | 62 total · 8 CRITICAL |

*Counts intentionally differ: labelled readings are dataset ground-truth episodes; breach readings are raw out-of-bounds values; detected events are spawned only after one metric stays out of bounds for 3 consecutive readings.*

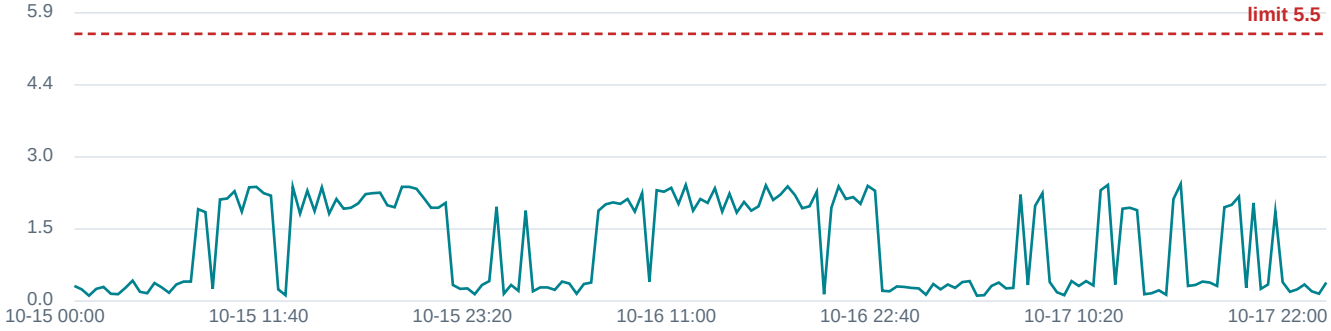
## SENSOR PERFORMANCE GRAPHS

### TEMPERATURE OVER TIME (°C)



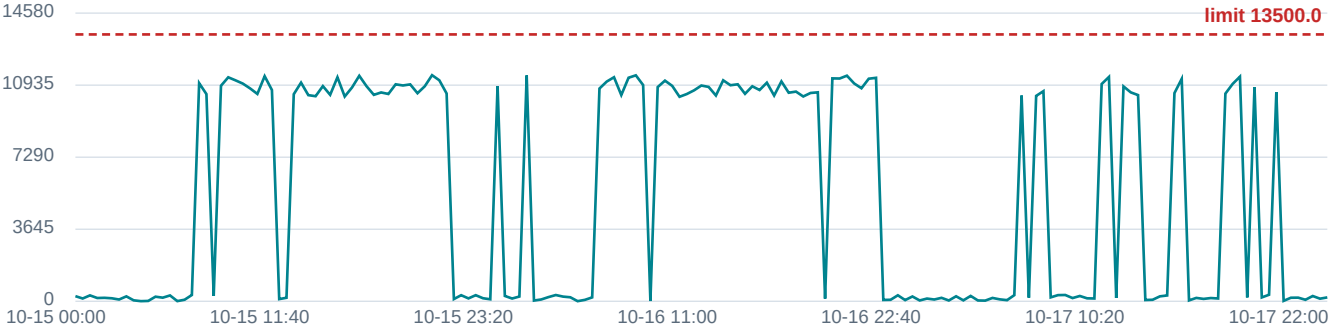

173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

### VIBRATION OVER TIME (MM/S)




173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

### MOTOR RPM OVER TIME (RPM)




173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

## ALERT HISTORY

| TIMESTAMP | METRIC | VALUE | THRESHOLD | SEVERITY | MESSAGE |
|---|---|---|---|---|---|
| 2026-10-16 15:15 | humidity_pct | 85.0 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 15:10 | humidity_pct | 81.4 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 15:05 | humidity_pct | 84.1 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 15:00 | humidity_pct | 81.5 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:55 | humidity_pct | 82.0 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:50 | humidity_pct | 82.6 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:45 | humidity_pct | 79.9 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:40 | humidity_pct | 80.4 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:35 | humidity_pct | 78.3 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:30 | humidity_pct | 79.1 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:25 | humidity_pct | 77.8 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:20 | humidity_pct | 78.0 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:15 | humidity_pct | 77.6 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:10 | humidity_pct | 76.6 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:05 | humidity_pct | 77.1 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 14:00 | humidity_pct | 75.2 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 13:55 | humidity_pct | 75.5 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 09:50 | humidity_pct | 84.0 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 09:45 | humidity_pct | 84.4 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 09:40 | humidity_pct | 83.5 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 09:35 | humidity_pct | 82.1 | 70.0 | WARNING | Humidity above limit |
| 2026-10-16 09:30 | humidity_pct | 81.8 | 70.0 | WARNING | Humidity above limit |

*Showing latest 22 of 62 alerts, newest first. The full history is available in alerts.csv.*

## SUMMARY & RECOMMENDATIONS

This printer shows an elevated labelled-anomaly rate of 8.7% against a fleet-wide 6.0%.

Power draw exceeded its limit on 14 reading(s) (limit 51.3 kW, max seen 66.3 kW). Recommend checking electrical load and supply stability.

Humidity fell outside the 30–70% band on 40 reading(s). Recommend reviewing site climate control.

Paper jams exceeded 2 per interval on 8 reading(s). Recommend checking feed rollers and media quality.

Note: recommendations are rule-based responses to threshold breaches present in the monitored data. They are maintenance suggestions, not confirmed diagnoses.